

Handling Class Imbalance Problem for Diabetic Retinopathy Fundus Image Classification

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Abstract—Diabetic Retinopathy (DR) is a leading cause of vision impairment, necessitating early detection through automated deep learning models. This study explores the use of pre-trained Convolutional Neural Networks (CNNs), specifically EfficientNet B5 and B7, for DR classification especially using imbalanced fundus image dataset from the Asia Pacific Tele-Ophthalmology Society (APTOS) 2019. A key challenge in this task is the class imbalance in DR severity levels, which can reduce model performance. To address this, we employ class weighting, L2 regularization, data augmentation, and upsampling techniques to enhance model generalization. Through extensive experimentation, EfficientNet B5 and B7 architectures were fine-tuned over multiple iterations to optimize performance. Results indicate that EfficientNet B7 achieved a slightly superior test accuracy of 93.1% compared to 92.7% for EfficientNet B5. Both models attained an overall F1-score of 92%, with B7 demonstrating better generalization by achieving the highest minimum class-wise F1-score. While EfficientNet B5 is more suitable for deployment in resource-constrained environments due to its lower batch size, B7 is better suited for large-scale applications. Comparative analysis with related studies reveals that dataset size and model architecture significantly impact performance. The findings underscore the potential of transfer learning and hyperparameter tuning in improving DR classification and early diagnosis, contributing to more accessible and efficient ophthalmic screening solutions.

Keywords—Diabetic Retinopathy Classification, Deep Learning, Convolutional Neural Network, Upsampling, Transfer Learning, Data Augmentation

I. INTRODUCTION

Deep learning, a specialized area within machine learning that uses algorithms to model complex patterns in data by employing multiple layers of processing to learn high-level representations, relying on intricate structures and non-linear transformations to analyze and interpret the data [1], has expanded the possibilities of tasks that can now be done in the field of Artificial Intelligence. Each layer is designed to progressively identify more intricate patterns and representations from the data inputted. These layers often involve nonlinear transformations, enabling the system to model intricate relationships that simpler methods might overlook [2]. In contrast to traditional machine learning approaches that require manually designed features, deep learning systems are capable of automatically learning representations from raw data. At the core of deep learning are artificial neural networks, particularly deep neural networks (DNNs), which consist of multiple interconnected layers that facilitate the extraction of complex patterns and insights. This layered architecture allows for a hierarchical approach to learning, where earlier layers capture basic features while deeper layers identify increasingly

sophisticated and abstract patterns. This approach has shown exceptional effectiveness in tasks such as image and speech recognition, classification, natural language processing, and other domains that involve complex data interpretation. Its rapid progress has been driven by advancements in computational power, the availability of large datasets, and the development of innovative neural network architectures, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers [3].

This research is focused on CNNs, which is a type of artificial neural network or a class of deep neural network designed to process structured grid-like data, such as images, by learning spatial hierarchies of features through convolutional filters, pooling operations, linear transformations, and activation functions, enabling efficient pattern recognition and classification [1]. The convolution operation, a hallmark of CNNs, involves applying kernels (filters) to highlight specific features within the input data. Each kernel learns to detect unique patterns, which are then passed through pooling layers to reduce spatial dimensions and computational complexity while preserving critical features. CNNs can either be pre-trained or custom; pre-trained are models that have been trained for a task but can be used for another task, while custom models are built and trained from scratch.

In this research, pre-trained CNN models like EfficientNet B5 and B7 will be used to classify fundus images obtained from Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 blindness detection competition [4]. Handling this data is particularly challenging due to the limited number of images available and the imbalance in the distribution of images across different DR classes.

Pre-trained models possess an advantage known as transfer learning which is a technique where a model trained for one task is used for another task. This requires using the feature extraction layers of the model, removing the original classification layers and adding a new one better suited for your task, and fine-tuning the hyper-parameters.

At this point, let us define Diabetic Retinopathy (DR). It is an eye disorder that develops when elevated blood sugar levels damage the tiny blood vessels in the retina, the layer at the back of the eye that is sensitive to light, which is responsible for capturing visual details.[5]. People living with diabetes are very likely to suffer from DR. It can cause blurring and spots in your vision, and if untreated, it may lead to loss of sight or blindness. Diagnosis relies on regular screening of fundus images, with the severity of the disorder classified into Non Proliferative DR and Proliferative DR stages; which are subdivided into Early/Mild NPDR,

Moderate NPDR, Severe NPDR and Proliferative DR [6]. These four subdivisions in addition to those who do not have DR (No DR) make up the five classes in our proposed data.

Recent epidemiological data from the American Academy of Ophthalmology reveals that the global prevalence of diabetes mellitus currently stands at 387 million, with projections indicating an increase to 592 million by 2035 [7]. This work aims to help in the fight to arrest this growth through early screening and detection using pre-trained models.

The objective of this work is to find the optimal pre-trained model for transfer learning that can learn features from fundus images. Through several experiments, it was found that EfficientNet models, especially B5 and B7 are suitable for DR classification. However, the class imbalance problem causes the biggest problem and it is the focus of our work. Therefore, the contributions of the work can be summarized as follows:

- Delivering a deep learning model based on transfer learning that can precisely predict the disease's severity in patients despite the class imbalance present in diabetic retinopathy data. Furthermore, the model will be computationally efficient.
- Handling class imbalance problem by increasing the number of samples or images in the minority classes to avoid the model being skewed in favour of the majority class (upsampling). In addition to upsampling, we applied class weighting, L2 regularization, data augmentation.

This paper's content is organized as follows: Section II presents relevant works on diabetic retinopathy classification, whereas Section III describes the dataset used in this work. The architectures of the pre-trained models and the proposed model are described and explained extensively in section IV; the evaluation and results are discussed in section V; and a conclusion is presented in section VI.

II. LITERATURE REVIEW

A review of the proposed models in related works showed that most researchers employed data preprocessing techniques such as CLAHE for contrast enhancement, Gaussian filtering for noise reduction, and various data augmentation strategies like flipping, rotation, and zooming – these processes helps produce a robust model as well as deal with class imbalance. Image input sizes varied, with some studies using as low as 224×224 pixels. The Adam optimizer was a common choice, along with the Softmax activation function for multi-classification tasks. Categorical Cross-Entropy was the most frequently used loss function. The APTOS dataset was widely used, making it a standard benchmark for DR classification. EfficientNet models frequently appeared, either as top-performing models or for comparison, while custom CNNs, ensemble models, and hybrid architectures consistently achieved high accuracy. Learning rates ranged from 0.001 to 0.00001, often adaptively adjusted, and batch sizes typically varied between 16 and 64, depending on model complexity. These insights provide a strong foundation for building an optimized DR classification model. More details about the related works are as follows:

Alwakid et al presented a specialized CNN model leveraging CLAHE and ESRGAN to enhance image quality [8]. The model achieved an impressive 97.83% accuracy, demonstrating the effectiveness of custom architectures with targeted preprocessing. However, their training/test split is 95/5, which is not a realistic scenario. The findings emphasize that while lightweight models improve efficiency, custom CNNs can outperform them when computational power is available.

Sait lightweight and efficient model designed for resource-constrained environments, MobileNet V3-Small utilized YOLO V7 for feature extraction and QMPA for feature selection [9]. It reached an accuracy of 98%, highlighting that computationally efficient models can still achieve high performance, making them suitable for real-world deployment in telemedicine.

Youldash et al. EfficientNet B3 showed strong performance in multi-class classification, achieving 85.1% accuracy [10]. Its use of compound scaling and contrast enhancement techniques underscores the importance of balancing efficiency and accuracy, particularly for detailed DR staging. The study also demonstrated the advantage of using different models for binary and multi-class classification tasks.

Majumder and Kehtarnavaz's multitasking model based on Xception architecture incorporated both classification and regression tasks, reaching 86% accuracy [11]. The integration of multitasking learning improved generalization, proving beneficial for complex medical imaging problems. However, its performance was slightly lower than single-task models, suggesting a trade-off between flexibility and accuracy.

Ghosh & Chatterjee's VGG16 + Inception V3 ensemble model combining two pre-trained networks, this approach leveraged feature diversity to achieve 96.4% accuracy [12]. The findings support the effectiveness of ensemble learning in boosting classification accuracy and reducing misclassification, particularly in severe DR cases.

Custom architectures, hybrid models, and ensembles tend to achieve the highest accuracy, especially when combined with advanced preprocessing techniques. Lightweight models such as MobileNet V3-Small offer practical deployment advantages without significant performance trade-offs. EfficientNet models consistently emerge as strong contenders, balancing accuracy and computational efficiency. These insights will guide the development of a robust DR classification model, ensuring an optimal balance between accuracy and efficiency.

III. DATASET

APTOS 2019 blindness detection competition data [4] obtained from Kaggle was used. The dataset originally contained 3662 images. Tools utilized for this study included Google Colab with an A100 GPU accelerator, Python 3.9, and Jupyter Notebook.

The dependencies used included TensorFlow 2.17.1 with the Keras library for model building, PIL and OpenCV for image processing, Multiprocessing for parallelization, OS for file handling, Scikit-learn for data splitting, NumPy for numerical computations, Matplotlib for visualization, Pandas for CSV handling, and Kaggle API for dataset access.

Data preprocessing involved trimming black borders, resizing images to 300×300 pixels while maintaining aspect ratio, and processing them into a new directory structure using multiprocessing for efficiency. These steps ensured consistency, minimized irrelevant data, and optimized images for model training.

In our work, to address class imbalance problem, the number of images in the ‘Mild,’ ‘Moderate,’ ‘Proliferative DR,’ and ‘Severe’ classes were upsampled to 800, 1200, 800, and 800, respectively. This was achieved by randomly selecting existing images, applying Gaussian noise, and slightly adjusting contrast to create new samples. This upsampling improved class distribution, enhancing model generalization and reducing bias. In particular, adding Gaussian noise has a significant effect on the model's ability to generalize features for underrepresented classes.

The final pre-processed dataset consisted of 5405 images across five classes. The data was split into training and validation, to test sets with an 85%-15% ratio. The resulting distribution was: 4594 training and validation images, and 811 test images as shown in Figure 1.

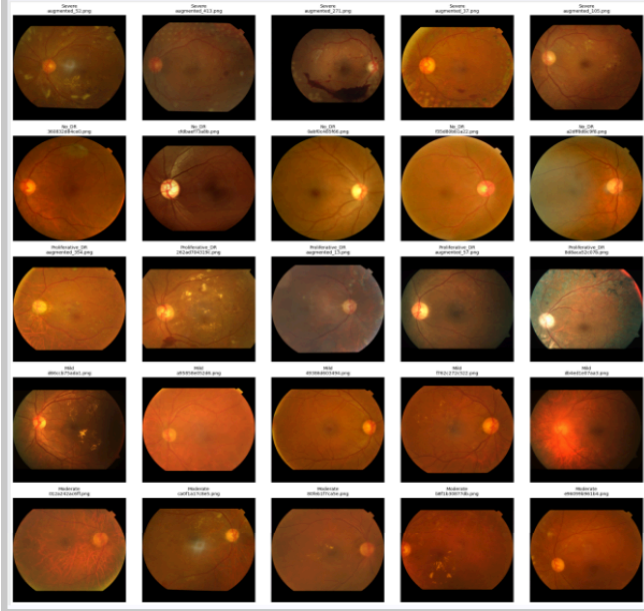


Fig. 1. APTOS preprocessed dataset sample images

TABLE I. DATASET DETAILS

APTOS Dataset (DR)	
Total Number of Original Samples	3662
Total Number of Samples after Preprocessing(Upsampling)	5405
Total Number of Classes	5
Train to Test Ratio	85% - 15%
Total Training Set	4594
Total Test Set	811

TABLE II. DISTRIBUTION OF SAMPLE IMAGES PER CLASS

	Original	Preprocessed
No DR	1805	1805
Mild	370	800
Moderate	999	1200
Severe	193	800
Proliferative DR	295	800

To enhance model performance and prevent overfitting, various data augmentation techniques were applied to the training dataset. The pixel values were normalized from a range of [0, 255] to [0, 1] by using a rescale factor of 1./255. Images underwent random rotations of up to 30 degrees, with horizontal and vertical shifts of 20% of their dimensions. Shear distortions with an intensity of 0.2 and zoom adjustments of up to 20% were incorporated. Additionally, random horizontal and vertical flips were implemented to introduce diversity in the dataset. The nearest-neighbor method was utilized to fill in missing pixel values caused by transformations. For the validation dataset, only rescaling with a factor of 1./255 was applied to standardize the pixel values while ensuring the integrity of the data remained intact and minimizing unnecessary alterations.

IV. PROPOSED DEEP TRANSFER LEARNING BASED ON CNN MODELS

A. EfficientNet B5 Architecture

With 303 layers and default input size 456×456 pixels, EfficientNet-B5 is a one of the powerful extensions of EfficientNet variants. It incorporates Mobile Inverted Bottleneck Convolutions (MBConv) along with squeeze-and-excitation (SE) blocks, which work together to enhance the model's ability to extract features effectively as shown in Figure 2 [13]. A key innovation of EfficientNet B5 is its compound scaling strategy, which balances the network's depth, width, and input resolution in a way that maximizes accuracy while minimizing computational costs. Its architecture strikes an optimal balance between computational cost and performance, making it an ideal choice for applications where both precision and efficiency are crucial.

It is suitable for larger datasets with greater accuracy compared to EfficientNet B3 and B4, and it efficiently handles intricate tasks. The downside is, it requires more computational resources than smaller variants and may be an overkill for tasks with limited complexity or smaller datasets.

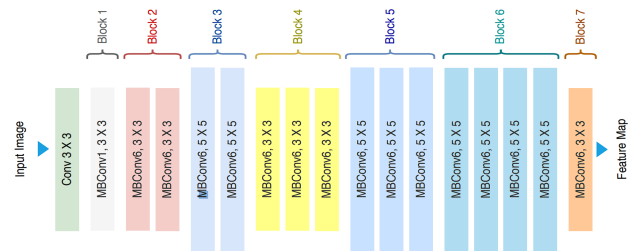


Fig. 2. EfficientNet (B5) Architecture

B. EfficientNet B7 Architecture

EfficientNet-B7, as shown in Fig 3 is the deepest model in the EfficientNet series with 466 layers and default input size 600×600 pixels, takes the compound scaling strategy to its maximum. By significantly increasing depth, width, and resolution, it can process high-resolution images and large-scale datasets with remarkable accuracy. This extended architecture allows it to capture intricate patterns

in data, making it well-suited for complex tasks that require detailed feature extraction [13].

This model delivers state-of-the-art accuracy for large-scale datasets such as ImageNet and excels in high-resolution and detail-oriented tasks like satellite image analysis. Weaknesses include high computational and memory requirements, necessitating advanced hardware such as TPUs or GPUs and challenges deploying it in resource-constrained environments.



Fig. 3. EfficientNet (B7) Architecture

C. Handling Class Imbalance

One of the main challenges of this research, as stated in the aims and objectives, is to deliver a model that can predict with precision the level of severity of DR using fundus images of patients despite the imbalance in the sample distribution of the data. For this reason, we experimented with class weighting, L2 regularization, data augmentation and upsampling;

Class Weight: Increases the importance of underrepresented classes in the loss calculation to ensure the model learns effectively from them [24].

L2 Regularization: Also called Ridge regularization, incorporates the squared sum of the model's coefficients into the loss function. That is, it adds a penalty for large weights to the loss function, reducing overfitting and improving the model's ability to generalize across all data, including imbalanced classes [24].

Data Augmentation: Enhances the variety and quantity of minority class samples during training by applying transformations to existing images, reducing bias and overfitting [25]. This process has been explained in the data overview.

Upsampling: Balances the dataset by replicating or creating new samples for underrepresented classes, ensuring fair training exposure [26]. This process has been explained in the data overview.

D. Transfer Learning and Fine-Tuning EfficientNet B5 & B7 Models for Diabetic Retinopathy Classification

The proposed models leverage transfer learning using EfficientNetB5 and EfficientNetB7 as base models, both pre-trained on ImageNet. The top classification layers are excluded, and the input shape is set to (300, 300, 3). These models are further refined with additional layers to enhance feature extraction and classification.

For their first experiments, both architectures incorporate a MaxPooling2D layer to reduce spatial dimensions, followed by a Flatten layer to convert the output into a one-dimensional vector. The EfficientNetB5 model features a single dense layer with 512 units, ReLU

activation, and L2 regularization (0.002), while the EfficientNetB7 model has multiple dense layers: 1024, 512, and 256 units, each with ReLU activation and L2 regularization. Dropout layers (0.5, 0.3, and 0.3) are applied in EfficientNetB7 to mitigate overfitting. Both models conclude with a softmax-activated output layer consisting of five units, corresponding to the five classification classes.

For optimization, both models use the Adam optimizer with a learning rate of 0.0001. The loss function is sparse categorical cross-entropy, making it suitable for multi-class classification. Accuracy is the key evaluation metric during training, and F1-score after testing, providing a measure of the models' performance in predicting diabetic retinopathy severity levels.

EfficientNet B5 and B7 hyperparameters were fine-tuned over seven and eight experiments respectively as shown in the Table III and IV below. In the following section, we discuss the results of these experiments that reveal the effect of different techniques that we apply for handling the class imbalance problem.

TABLE III. EfficientNet B5 seven experiments

No	Batch size	Epoch	Class Weight	Dense Layers/neuron	Dropout	L2
1	8	50	No	1;512	No	0.02
2	16	50	No	1;512	No	0.02
3	32	50	No	1;512	No	0.02
4	8	50	Yes (loss)	1;512	No	0.02
5	8	50	Yes (loss)	1;512	No	0.01
6	8	50	Yes (data aug)	1;512	No	0.01
7	8	50	Yes (loss & data aug)	1;512	No	0.01

TABLE IV. EfficientNet B7 eight experiments

No	Batch size	Epoch	Class Weight	Dense Layers/neuron	Dropout	L2
1	4	50	No	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.02, 0.01, 0.005
2	8	50	No	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.02, 0.01, 0.005
3	16	50	No	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.02, 0.01, 0.005
4	32	50	No	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.02, 0.01, 0.005
5	32	50	Yes(loss)	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.02, 0.01, 0.005
6	32	50	Yes(loss)	3;1024, 512, 256	3;0.5, 0.3, 0.3	3;0.01, 0.005, 0.0025
7	32	50	Yes(loss)	1;1024	No	1;0.01
8	32	50	Yes(loss)	1;1024	No	1;0.001

V. EVALUATION AND RESULTS

A. Evaluation Setup and Metrics

Evaluation metrics play a vital role in assessing the effectiveness of machine/deep learning models. They

provide numerical values that help evaluate how well a model is performing on a specific task. These metrics can vary depending on the type of problem, such as classification, regression, or clustering [14]. In classification tasks, common metrics include accuracy, precision, recall, and F1-score, each serving a distinct purpose. Selecting appropriate metrics ensures a comprehensive evaluation, particularly in cases with imbalanced datasets where some metrics may give a skewed picture of performance [15].

When a model predicts the class of a sample, this prediction can be categorized as one of the following: *True Positive (TP)*: Instances where the model correctly predicts the positive class. *False Positive (FP)*: Instances where the model incorrectly predicts the positive class when the actual label is negative. *True Negative (TN)*: Instances where the model correctly predicts the negative class. *False Negative (FN)*: Instances where the model fails to predict the positive class and instead predicts it as negative. The number of samples under these categories are then used to calculate each metric.

Accuracy measures the proportion of correct predictions among all predictions made by the model [16]. While it is a straightforward and widely used metric, it can give misleading results when applied to datasets with class imbalances [17]. For example, if one class dominates the dataset, the model may achieve high accuracy simply by always predicting the majority class, even if it performs poorly on the minority class.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Precision evaluates the model's ability to make accurate positive predictions, by measuring the fraction of correctly identified positive cases (TP) out of all predicted positives (TP+FP) [18]. This metric is particularly important in scenarios like spam detection or medical diagnostics, where minimizing false positives is critical [19]. A high precision score indicates fewer irrelevant results in the positive predictions.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

Recall, sometimes referred to as sensitivity or the true positive rate, measures how well the model identifies all actual positive cases in the dataset. It is the fraction of true positives (TP) among all actual positive instances (TP+FN) [20]. Recall is crucial in applications where missing a positive case (a false negative) carries serious consequences, such as diagnosing diseases [21]. However, a high recall often results in lower precision, as the model may cast a wider net to capture more positives.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

The F1-score combines precision and recall into a single metric, using their harmonic mean to strike a balance between the two. This metric is particularly valuable when dealing with datasets that are not evenly distributed between classes - imbalance data [22]. Unlike accuracy, the F1-score ensures that both false positives and false negatives are given appropriate weight. It's ideal for use cases where both precision and recall are equally important [23].

$$F1 - Score = \frac{2*precision*recall}{precision+recall} \quad (4)$$

B. EfficientNet B5 Results

As shown in TABLE V, the highest validation accuracy, 92.7%, was recorded by experiment #1 and #4, but this did not translate to the test accuracy as experiment five achieved the highest test accuracy, 92.7%. In addition to that, experiment five had a validation accuracy of 91.4%, lower than its test accuracy, which means it performed even better on unseen data. Experiment #5 also has the best lowest f1-score, by class, of 88% and a joint highest f1-score of 99% with the others, making it the model that generalized the best. In experiment #5, a batch size of 8 was applied together with class weight and L2 regularization of 0.01 (see Table III). Overall Precision, Recall and F1-score results of Efficient B5 variants that we fine-tuned are also shown in Table VI. Table VI, shows that experiment five has the best overall precision (92%), recall (92%), and f1-score (92%) compared to others. These few points make experiment #5 the best B5 model. Confusion matrix of this best model is also shown in Figure 2. It is demonstrated that the moderate class was one of the challenging classes to handle.

TABLE V. EfficientNet B5 seven experiments results

No	Val Loss	Valid Acc (%)	Test Loss	Test Acc (%)	Lowest F1-Score (Class)	Highest F1-Score (Class)
1	0.3472	92.7	0.3931	92.1	0.85 (Moderate)	0.99 (No DR)
2	0.3767	91.4	0.4208	91.7	0.84 (Moderate)	0.99 (No DR)
3	0.3714	92.1	0.4225	90.6	0.82 (Moderate)	0.99 (No DR)
4	0.3417	92.7	0.4175	91.6	0.86 (Moderate)	0.99 (No DR)
5	0.4322	91.4	0.4094	92.7	0.88 (Moderate)	0.99 (No DR)
6	0.398	90.8	0.4272	91	0.84 (Moderate)	0.99 (No DR)
7	0.3948	92	0.4353	91.3	0.85 (Moderate)	0.99 (No DR)

TABLE VI. EfficientNet B5 seven experiments overall results

No	Overall Precision	Overall Recall	Overall F1-Score
1	0.91	0.91	0.91
2	0.91	0.91	0.91
3	0.89	0.9	0.89
4	0.9	0.91	0.9
5	0.92	0.92	0.92
6	0.9	0.9	0.9
7	0.9	0.9	0.9

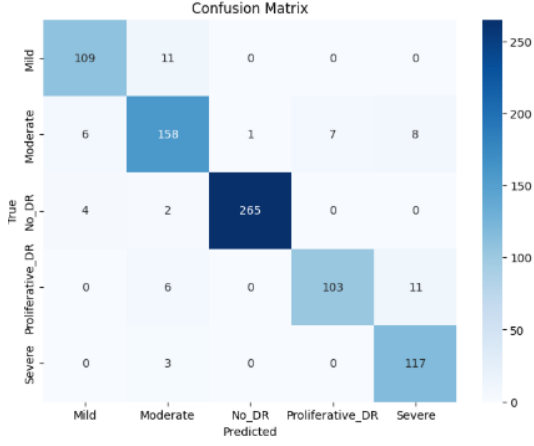


Fig 3. EfficientNet B5 experiment #5 confusion matrix

C. EfficientNet B7 Results

As shown in TABLE VII, the highest validation accuracy, 92.8%, was recorded by experiment #4, which also has the highest test accuracy, 93.1% – which means it performed even better on unseen data. Experiment #4 also has the best lowest f1-score of 89% and a joint highest f1-score of 99% with the others, making it the model that generalized the best. Experiment #4 has a larger batch size of 32 compared to EfficientNet B5 version’s batch size of 8. In addition, we have three dense layers with regularization and dropouts as shown in Table IV.

TABLE VIII, shows that experiment #4 has the best overall precision (92%), the joint best overall recall (92%), and the best overall f1-score (92%) compared to others. These few points make experiment #4 the best B7 model. In addition, for the best model confusion matrix is shown in Figure 4. It is shown that Moderate, No DR and Proliferative classes can be classified better by the EfficientNet B7 model. However, the Mild and Severe classes are better classified by the EfficientNet B5 model.

TABLE VII. EfficientNet B7 eight experiments results

No	Val Loss	Val Acc (%)	Test Loss	Test Acc (%)	Lowest F1-Score (Class)	Highest F1-Score (Class)
1	0.4713	90.4	0.4733	90	0.84 (Moderate, Proliferative DR)	0.99 (No DR)
2	0.4247	92.7	0.4398	91.7	0.87 (Moderate, Proliferative DR)	0.99 (No DR)
3	0.4518	93	0.5598	91.4	0.86 (Moderate)	0.99 (No DR)
4	0.5499	92.8	0.5281	93.1	0.89 (Moderate)	0.99 (No DR)
5	0.5819	92.1	0.6176	91.7	0.85 (Moderate)	0.99 (No DR)
6	0.5665	92.7	0.5972	90.4	0.85 (Moderate, Proliferative DR)	0.99 (No DR)
7	0.4218	91.6	0.3947	92.4	0.88 (Moderate, Mild)	0.99 (No DR)
8	0.4746	92.1	0.4243	92.4	0.87 (Moderate)	0.98 (No DR)

TABLE VIII. EfficientNet B7 eight experiments overall results

No	Overall Precision	Overall Recall	Overall F1-Score
1	0.88	0.89	0.88
2	0.91	0.91	0.91
3	0.9	0.91	0.9
4	0.92	0.92	0.92
5	0.9	0.91	0.91
6	0.89	0.89	0.89
7	0.91	0.92	0.91
8	0.91	0.92	0.91

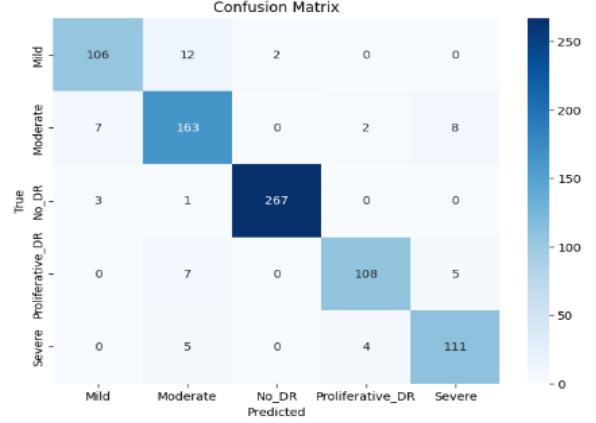


Fig 4. EfficientNet B7 experiment four confusion matrix

D. Computational Efficiency

The best EfficientNet B5 model which has a total parameters of 163,847,528 (625.03 MB) produced a model of 1,041.28 MB or 1.02 GB. Trained with a batch size of 8, this model is suited for resource constrained environments.

The best EfficientNet B7 model which has a total parameters of 390,254,920 (1.45 GB) produced a model of 2.42 GB. Trained with a batch size of 32, this model is suited for a more resourceful environment than B5.

E. Ablation Study

An ablation study was carried out on the best model, Efficientnet B7 model, to understand the impact of the upsampling process done during the data preprocessing, data augmentation process and regularization has on the performance of the model. TABLE X below shows the upsampling state and augmentation processes, L2 regularization experimented on, and their evaluation. Every setting and steps taken to prepare the data and train the model for B7 experiment #4, which is the best B7 model, remained the same at all times except for the processes contained in the Class Imbalance Handling column of Table IX.

The ablation study underscores the impact of upsampling, data augmentation, and regularization on model performance – without them, validation and test accuracy remain low (75.4% and 76.6%), with a poor F1-score of 0.21 for Mild cases. Upsampling alone significantly improves accuracy (91.2% validation, 91.9% test) and raises

the lowest F1-score to 0.86 for Mild and Moderate cases, proving its effectiveness in addressing class imbalance. Data augmentation alone yields moderate gains (82.8% validation, 83.5% test, F1-score 0.45), while L2 regularization alone performs the worst (72.1% validation, 70.4% test, F1-score 0.06 for Severe cases). Upsampling has the most substantial impact. Each augmentation technique was tested with upsampling and regularization. Among them, height shifting produced the highest test accuracy (93.1%) and F1-score (0.92), while vertical flipping and shear transformations resulted in lower performance (88.0% and 87.9% test accuracy, respectively). This suggests that moderate spatial transformations aid generalization, whereas extreme distortions like shear and vertical flips may introduce inconsistencies in learning.

TABLE IX. EFFICIENTNET B7 EXPERIMENT #4 ABLATION STUDY

Class Imbalance Handling	Val Acc (%)	Test Acc (%)	Lowest F1-score (Class)	Highest F1-score (Class)	Over. Prec.	Over. Rec.	Over. F1-score
No upsampling, rescale=1./255, No L2	75.4	76.6	0.21 (Mild)	0.97 (No DR)	0.62	0.57	0.55
Upsampling, rescale=1./255, No L2	91.2	91.9	0.86 (Mild, Moderate)	0.99 (No DR)	0.91	0.91	0.91
No upsampling, All augment, No L2	82.8	83.5	0.45 (Severe)	0.99 (No DR)	0.74	0.62	0.66
No upsampling, rescale=1./255, L2	72.1	70.4	0.06 (Severe)	0.92 (No DR)	0.49	0.45	0.45
Upsampling, rescale=1./255, L2	92.2	90.8	0.82 (Mild)	0.99 (No DR)	0.89	0.9	0.89
Upsampling, rescale=1./255, rotation range=30, fill mode = nearest, L2	90.1	88.7	0.80 (Mild)	0.94 (No DR)	0.89	0.87	0.88
Upsampling, rescale=1./255, width shift range=0.2, fill mode = nearest, L2	91	90.8	0.81 (Mild)	0.98 (No DR)	0.9	0.9	0.9
Upsampling, rescale=1./255, height shift range=0.2, fill mode = nearest, L2	92.5	93.1	0.87 (Mild)	0.99 (No DR)	0.92	0.92	0.92
Upsampling, rescale=1./255, shear range=0.2, fill mode = nearest, L2	87.3	87.9	0.81 (Mild, Moderate)	0.97 (No DR)	0.88	0.86	0.86
Upsampling, rescale=1./255, zoom range=0.2, fill mode = nearest, L2	91.1	90.6	0.79 (Mild)	0.98 (No DR)	0.9	0.89	0.89
Upsampling, rescale=1./255, horizontal flip=True, L2	90.8	91.3	0.83 (Mild)	0.99 (No DR)	0.91	0.89	0.9
Upsampling, rescale=1./255, vertical flip=True, L2	87.3	88	0.81 (Moderate)	0.98 (No DR)	0.86	0.86	0.86

F. EfficientNet B5, B7 and Related Works Results

Although both fine-tuned proposed models have the same overall precision, recall and f1-score, EfficientNet B7 has a slightly better test accuracy of 93.1% compared to 92.7% for B5. It also generalized slightly better, with its lowest F1-score being 89% compared to 88% for B5. Finally, B5 was trained with a batch size of 8, which makes it fit for deployment in resource constrained environments, while B7 was trained with a batch size of 32 making it fit for larger scale applications.

Comparing both proposed models to the models reviewed in this study, where the researchers worked with APTOS data, it is observed that [8], [9], and [12] models performed better on all evaluation metrics as shown in Table X. This might be as a result of the size of the datasets, as these three models each have a total of 10458, 5590, 6034 images respectively compared to 5405 for ours. In addition, while [10], and [11] models performed poorer with each having the same total of 3662 images and the proposed approach performs better than them. which is less than ours. It is important to note that this is not the sole reason why a model may perform less or more efficiently. For example, Youldash et al [10] used Efficientnet B3 which is not as good as B5 or B7 at extracting complex features. Other factors such as hyperparameter tuning, and preprocessing techniques, also contribute to overall performance.

TABLE X. PROPOSED EFFICIENTNET B5, B7 & RELATED WORKS COMPARISON TABLE

Ref	Model	Total Images	Train-Test Split	Acc (%)	Lowest F1_score (Class)	Prec.	Rec.	F1-score
[8]	Custom CNN	10458	95:5	97.83	0.92 (Severe)	0.98	0.98	0.98
[9]	MobileNet V3-Small	5590	70:30	98	0.92 (Mild)	0.93	0.94	0.937
[10]	EfficientNet B3	3662	80:20	85.1	0.47 (Severe)	0.85	0.85	0.844
[11]	MXception	3662	90:10	86	N/A	0.77	0.7	0.73
[12]	VGG16 + Inception V3	6034	80:20	96.4	0.90 (Moderate)	0.96	0.96	0.96
This study	EfficientNet B5	5405	85:15	92.7	0.88 (Moderate)	0.92	0.92	0.92
This study	EfficientNet B7	5405	85:15	93.1	0.89 (Moderate)	0.92	0.92	0.92

VI. CONCLUSIONS AND FUTURE WORK

This work presented how to handle imbalance class problem using transfer learning and different techniques like regularization, data augmentation and upsampling for Diabetic Retinopathy (DR) classification from fundus images. First, pre-trained Convolutional Neural Networks (CNNs) were explored, specifically EfficientNet B5 and B7,

for DR classification using an imbalanced fundus image dataset. Generating additional images for the minority classes helped in handling the class imbalance in both EfficientNet B5 and B7 models. This helped the models generalize better rather than learning the features of the majority class. Adding class weights further improved the performance of B5 but not B7.

Understanding the limitations of the upsampling technique used for minority classes can help refine it for improved accuracy. Future studies could explore the effects of a higher standard deviation for Gaussian noise, a broader contrast range, or additional samples for further class balancing. EfficientNet B7 may require longer training to reach peak accuracy. Increasing the batch size for B7 could also enhance performance, but this might necessitate more training data, particularly for minority classes. Applying this data preprocessing strategy to ensemble models, transformer-based architectures, or hybrid models could yield even better results.

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